Roll No.: _____

Amrita Vishwa Vidyapeetham

Integrated MSc Degree(Supplementary) Examination – July 2025

Second Semester

Common to

Int MSc Mathematics with MDS/Int MSc Physics with MSC

22MAT113 Multivariable Calculus

Time : 3 hrs | Maximum Marks:100

Answer all Questions

Qn No.		Marks	CO	Blooms Taxonomy level
1	a) Evaluate the limit $\lim_{(x,y)\to(0,0)} \left(\frac{x^3-y^3}{x^2+y^2}\right)$ by two path test b) Examine the continuity at the origin $f(x,y) = \begin{cases} \frac{x^2-y^2}{x^2+y^2}, & x \neq 0,\ y \neq 0 \\ 3, & x = 0,\ y = 0 \end{cases}$	(5+5)	CO1	L1
2	a) Find all local maximum, local minimum, saddle points for the function $f(x,y) = x^2 + xy + y^2 + 3x - 3y + 4$ b) Find the linearization of $f(x,y) = x^2 - xy + 3\,sinz$ at the point (2,1, 0). Find also the error incurred in the rectangle R given by $\lvert x-2\rvert \leq 0.01,\ \lvert y-1\rvert \leq 002,\ \lvert z\rvert \leq 0.01.$	(5+5)	CO1	L3
3	a) Find $\frac{\partial w}{\partial r}, \frac{\partial w}{\partial s}$, where $w = x^2 + y^2$, $x = r - s$, $y = r + s$ b) Determine the tangent plane and normal plane to the surface $f(x,y,z) = x^2 + y^2 + z - 9 = 0$ at the point (1, 2, 4)	(5+5)	CO1	L1

4	a) Find the directional derivative of $f(x,y,z) = 2xy + z^2$ at the point (1, -1, 3) in the direction of $i + 2j + 2k$. b) In what direction from (2, 1, -1) is the directional derivative of $g(x,y,z) = x^2yz^3$ is maximum? What is the magnitude of its maximum?	(5+5)	CO1,CO2	
5	a) Evaluate the double integral $\iint_R (x^2 + y^2)dx\,dy$, where R is the region bounded by $y = x$ and $y^2 = 4x$. b) Sketch the graph of the region R and hence evaluate the double integral $\int_0^a \int_0^{bx/a} x\,dy\,dx$ by changing the order of integration.	(5+5)	CO2	L2
6	a) Determine the curl and divergence of the vector $e^{xyz}(i + j + k)$ at the point (1, 2, 3). b) Determine whether the force field $\overline{F} = 2xzi + (x^2 - y)j + (2z - x^2)k$ is conservative or not .	(5+5)	CO2	L3
7	a) If $\overline{F} = (4xy - 3x^2z^2)i + 2x^2j - 2x^3z\,k$. Prove that the line integral $\int_c F.dr$ is independent of path , find the scalar potential $f(x,y,z)$ and hence determine the work done in moving an object from (1, 1, 1) to the point (0,0,0). b) Evaluate the work integral $\int_c xy^3ds$ along C, where C is the segment of the line $y = 2x$ In the xy plane from $A(-1,-2,0)$ to $B\ (1,2,0)$.	(5+5)	CO3	L2
8	a) Use Greens theorem, evaluate the line integral $\int_c (x^2 + xy)dx + (x^2 + y^2)dy$, where C is a square formed by $x = \pm 1, y = \pm 1$. b) Change the cartesian integral into polar integral $\iint_R x\,dx\,dy$, where R is the region bounded by the unit circle in the first quadrent.	(5+5)	CO3,CO4	L3
9	a) Determine the surface area of the part of $z = xy$ that lies inside the cylinder $x^2 + y^2 = 1$. b) Evaluate the surface integral through the surface whose parametric representation is	(5+5)	CO5	L4

	$S; r(u,v) = u\,i + v\,j + (2u + 3v)k$, $0 \le u \le 2, -1 \le v \le 1$, if the velocity vector field is $F = 3x^2 i + y^2 j$.			
10	a) Evaluate the triple integral $\int_0^1 \int_0^{1/2} \int_{-2}^{2} (sin\pi x\, cos\pi y + 2)\, dz\, dy\, dx$ b) State Gauss divergence theorem using this theorem evaluate the surface integral $\iint_S F.n\, dA$, where the vector field $F = cosy\, i + sinx j + cosz\, k$ and S is the surface of $x^2 + y^2 \le 4,\ |z| \le 2$.	**(5+5)**	**CO4,CO5**	**L5**

L_1-Remember, L_2-Understand, L_3-Apply, L_4-Analyze, L_5-Evaluate, L_6-Create